

CTA Strategy · Course Notes

Harry Chen

Contents

1 · What Is a CTA Strategy	3
1.1 What a CTA Trades	3
1.1.1 Index, ETF, or future	3
1.1.2 Why futures rather than ETFs	5
1.2 Momentum, the Source of the Return	5
1.2.1 What momentum is	5
1.2.2 Why momentum might work	5
1.3 Long and Short, the Two Sides of a Position	6
1.3.1 What the two sides mean	6
1.3.2 How a short is borrowed	6
1.3.3 Why the short leg is the fragile one	6
1.3.4 How a short is represented	7
1.4 Who Else Is Trading	8
Background	8
Appendix · Notation	9
2 · Testing a Signal	11
2.1 A signal is a hypothesis, not a formula	11
2.1.1 The three objects	11
2.1.2 Why studying the signal is the same as studying the portfolio	12
2.2 What you hoped for, and what you get	14
2.2.1 Which return the signal is scored on	14
2.2.2 The plot everyone draws first	15
2.2.3 Why the real one is a cloud	15
2.3 Two ways out of the cloud	17
2.3.1 Turn down alpha	17
2.3.2 Beta Method	18
2.3.2.1 What a bar plot is	18
2.3.2.2 How the bar plot shows a trend	20
2.3.2.3 Is the staircase real?	22
2.3.2.4 What the bar plot cannot show	24

2.4	Step 2, corrected — risk-adjusted momentum	25
2.5	What to report — and why it is not a Sharpe ratio	27
2.6	Information availability	28
	Background	28
	Appendix · Notation	29

5 · Modelling 31

5.1	Why one split is not a backtest	31
5.1.1	The textbook split	31
5.1.2	What the single split costs	31
5.1.3	The fix that is not one	32
5.2	The walk-forward ladder	33
5.2.1	How long a segment should be	33
5.2.2	The slide	33
5.2.3	One rung, in full	33
5.3	Two validation layers	34
5.3.1	Why the inner layer cannot be skipped	35
5.3.2	What only the outer block may decide	35
5.3.3	The optimizer's parameters	36
5.4	Preparing a signal for a model	36
5.4.1	Clipping: keep the observation, cap the value	36
5.4.2	Standardizing: put every signal on one scale	38
5.4.3	Which mean and standard deviation you may use	39
	Appendix · Notation	39

1 · What Is a CTA Strategy

1.1 What a CTA Trades

Definition 1.1 (CTA strategy). A *CTA* (*Commodity Trading Advisor*) strategy is a rule-based strategy that trades **futures** systematically across asset classes—equities, fixed income, commodities, currencies—targeting **absolute return**, independent of market direction.

Note. The name is a historical artifact. Modern CTAs are not commodity-only; typical markets are equity-index futures, Treasuries, FX, energy, metals, and agriculture.

1.1.1 Index, ETF, or future

“S&P 500” names three different objects. Only two of them can be traded.

Layer	Example	What it is	Tradeable
Index	SPX, NDX	A published number computed from its constituents	No
ETF	SPY, VOO, QQQ	A fund holding the constituents; its own shares trade	Yes
Future	ES, NQ	A contract to settle at a future date; holds nothing	Yes

Definition 1.2 (Index).

$$I_t = \frac{M_t}{D_t}, \quad M_t = \sum_i N_i P_{it}$$

where P_{it} is the price of constituent i , N_i its float-adjusted share count, M_t the total float capitalisation, and $D_t > 0$ the *divisor*.

Claim 1.3. Index returns are the capitalisation-weighted average of constituent returns; the divisor sets only the level.

Proof. Both halves follow from I_t being a *sum divided by a constant*. A constituent’s dollar move is N_i times its price change, so its share of the index move is its share of total capitalisation:

$$r_I = \sum_i w_{it} r_i, \quad w_{it} = \frac{N_i P_{it}}{M_t}, \quad \sum_i w_{it} = 1.$$

And D_t divides I_t at both dates, so it cancels from the ratio—it can shift the level anywhere without touching a single return. $\square$

Note (Two consequences). **The top names dominate.** At $w \approx 0.07$ against $w \approx 0.0002$, the largest constituent carries roughly **350 times** the index impact of the smallest; this is not an average of 500 *prices*. **And the level is arbitrary.** D_t cancels from returns, so the scale is whatever the base period set it to—for the S&P 500, $1941-43 = 10$. **Compare returns, never levels.**

A price level is meaningful only against that same series' own history. Across tickers it carries no information at all—what each ETF's share price happens to be was fixed by how the fund was structured at inception. Dividing every series by its own first observation removes that arbitrariness:

Ticker	Raw close, start	Raw close, end	Rebased	Return
SPY	325.05	740.83	227.9	+127.9%
TLT	137.10	87.35	63.7	−36.3%
GLD	143.97	368.49	255.9	+155.9%

SPY has the highest closing price on every single day of the sample, yet GLD returned more. The end level alone cannot tell you this; you need the start it is measured against.

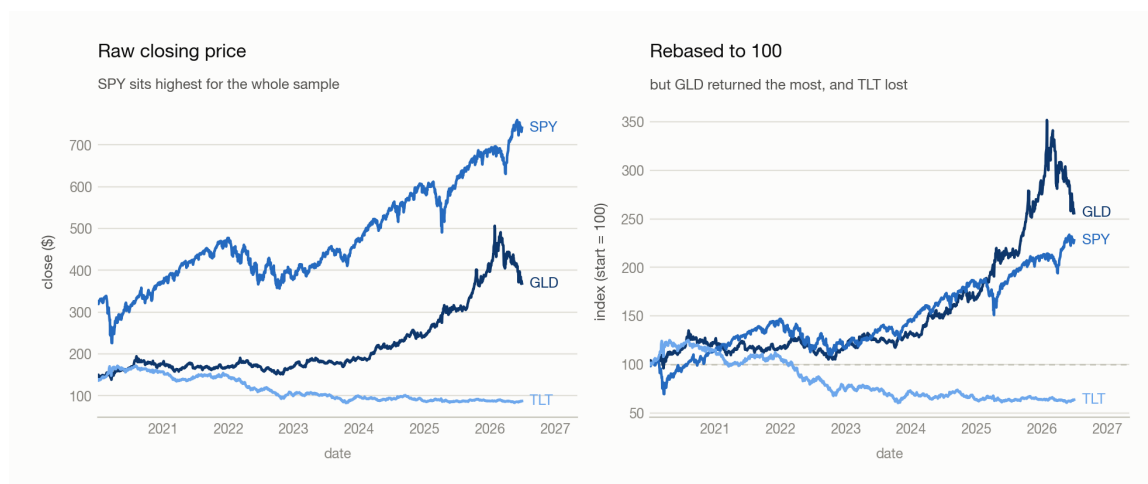


Figure 1.1: The same three ETFs, twice. Left, raw closing prices: SPY runs highest for the entire sample. Right, each series divided by its own first value and multiplied by 100: GLD now finishes highest and TLT ends below the line it started on. The ordering of the levels and the ordering of the returns are different orderings.

1.1.2 Why futures rather than ETFs

Reason	Futures	ETFs
Symmetry	Shorting costs nothing extra	Shorting needs borrowed shares, a fee, availability
Coverage	Deep markets in all four asset classes	Poor in commodities and FX
Capital efficiency	Margin is roughly 5 to 10 percent of notional; spare cash earns collateral yield	Full notional tied up

Symmetry is decisive: a strategy that must go short as freely as long cannot tolerate the asymmetry, and §1.3.3 shows why the short leg is the fragile one. The cost is that futures expire, so a continuous series has to be stitched together from contracts—see 100 · The Dataset.

1.2 Momentum, the Source of the Return

1.2.1 What momentum is

Definition 1.4 (Momentum). *Momentum* is the empirical tendency for recent relative performance to persist: assets that outperformed over the past one to twelve months tend to keep outperforming, and recent laggards tend to keep lagging.

Note. The claim is about *returns*, not price levels—it concerns the ordering of performance across assets and across time. That is why one momentum rule can be applied to markets with nothing else in common, from Treasury futures to natural gas.

Turning this into a number—a lookback window, a normalization, a test that it carries information—is 2 · Testing a Signal’s subject. Here it is only the premise.

1.2.2 Why momentum might work

Two contested explanations, no proof. Momentum has been durably profitable regardless.

- **Information diffusion.** Unglamorous macro news spreads gradually, creating sustained pressure. Possibly weakened since 2008, as computing and data distribution got faster.
- **Selective momentum capture.** Winners keep winning and losers keep losing, because large positions take time to build and unwind, investors chase winners, and major events do not resolve in a single day.

2 · Testing a Signal turns this from a story into something testable.

1.3 Long and Short, the Two Sides of a Position

1.3.1 What the two sides mean

Definition 1.5 (Long). Buy first, sell later. Profit is the sell price minus the buy price.

Definition 1.6 (Short). Borrow and sell first, buy back and return later. Profit is the sell price minus the buy-back price.

Note. Running both deploys capital fully—a 150/50 or 200/100 book rather than idle cash—and lets the strategy express a negative view, which a long-only book cannot.

1.3.2 How a short is borrowed

A short is not selling out of thin air—the shares are borrowed before they are sold:

1. **Borrow** N shares from a lender—a broker, or a long-term holder.
2. **Sell** them now, and receive cash.
3. **Buy back** N shares later, from the market.
4. **Return** the N shares, closing the position.

Note. You never create shares. They are fungible, so returning “ N shares of SPY”—not specific certificates—settles the debt. **Lenders are paid.** Long-term holders earn a **lending fee** on stock they were going to sit on anyway. **And the broker sits in the middle**, matching the two sides and holding margin; the market for this is **securities lending**.

→ The full mechanics beneath this—who holds the shares mid-loan, who receives the dividend, hard-to-borrow rates, and how a short squeeze runs away: §10 of the `market-101-foundations` note in `market_knowledge/`.

1.3.3 Why the short leg is the fragile one

Claim 1.7. For a static position, a long’s loss is bounded by its outlay; a short’s is not.

Proof. With signed share count s , entry price $P_0 > 0$ and exit price P_T , the payoff is

$$\text{PnL}(P_T) = s(P_T - P_0), \quad P_T \in [0, \infty)$$

the domain being half-open because a price cannot go negative.

For a **long**, $s > 0$: PnL is increasing in P_T , so its infimum is at the left endpoint, $\text{PnL} \geq s(0 - P_0) = -sP_0$ —exactly the amount paid. For a **short**, $s < 0$: PnL is decreasing in P_T , and the domain has no right endpoint, so $\text{PnL} \rightarrow -\infty$ as $P_T \rightarrow \infty$. □

The asymmetry is in the **domain**, not the payoff—both are lines of slope $|s|$. The floor exists only because prices are floored at zero.

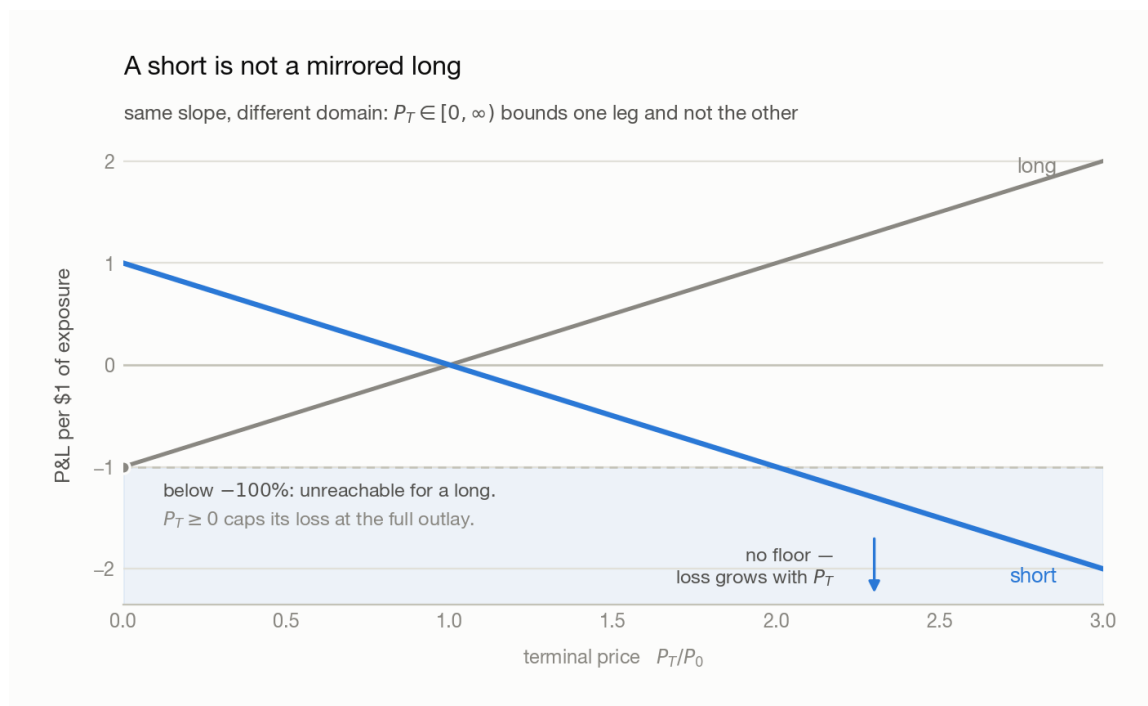


Figure 1.2: Profit and loss per dollar of exposure against the terminal price P_T . The long and short payoffs are lines of equal and opposite slope crossing at the entry price P_0 ; the long's line stops at -100 percent where the price reaches zero, while the short's continues downward without limit as the price rises. Illustrative.

Note. Hence margin requirements and forced buy-ins—and why a 150/50 book, 200 percent gross, is **two times leverage** rather than free money.

1.3.4 How a short is represented

A short needs no separate machinery. Carry the position as a **signed quantity** and let the sign do the work: a negative share count valued at the market price is a **negative position value**, which rises exactly when the price falls.

Note (Sign convention). A negative position value is a **liability**—a debt owed—not a loss already taken.

Note (Static versus constant-exposure shorts). The proof above holds the share count fixed at entry. A book that instead maintains a constant *dollar* exposure buys the short back down as it moves against it, so the unbounded loss is never realised. That daily rebalancing is an **implicit risk control**: the proof is right, but a constant-exposure book is not running the position it describes.

→ How this project implements it, with the accounting derivation and the measured gap between the two: `Backtests.md` in `Backtest_prototype/`.

1.4 Who Else Is Trading

Participant	Examples	Typical hold
Index / passive funds	Vanguard, BlackRock	Years to decades
Active managers	Fidelity, PIMCO	Monthly to quarterly
Hedge funds	—	Intraday to several days
Market makers	Optiver, Citadel Securities, IMC, SIG	Seconds to minutes
Noise traders	Retail, uninformed flow	No consistent horizon

Background

General finance knowledge needed to read this chapter, not part of its argument.

What is the relationship between futures/options and asset classes?

They answer different questions, so they are not alternatives to one another.

- **Asset class**—*what* underlying market you are exposed to.
- **Instrument**—*how* you hold that exposure: cash, futures, or options.

Futures are not an asset class, and options are not a rival one. Every asset class trades in every instrument:

Asset class	Typical underlying	Futures	Options
Equities	S&P 500, Nasdaq 100	ES, NQ	options on ES
Fixed income	US Treasuries, Bunds	ZN (10-year), ZB (30-year)	options on ZN
Commodities	Crude oil, gold, corn	CL, GC, ZC	options on CL
Currencies (FX)	EUR/USD, USD/JPY	6E, 6J	options on 6E

The same exposure comes in three wrappers: equity exposure is bought in cash as SPY, in futures as ES, or in options as an ES call. So in the opening definition—“trades **futures** systematically **across asset classes**”—the three parts are independent: **futures** is the wrapper, **across asset classes** is the breadth of underlying markets, and **systematically** means by rule or model rather than discretion.

Which asset class does a bond belong to?

Fixed income. Treasuries, corporate, municipal and high-yield bonds all sit there.

- “Fixed” describes the **contract**, meaning a scheduled coupon and the return of principal—not a fixed price.
- Bond prices still move, with interest rates, credit risk and time to maturity. That variation is what makes them a trend market at all.
- Treasury futures trade at 2-, 5-, 10- and 30-year maturities, so a CTA picks the *maturity* it wants exposure to.

Why do CTAs use futures rather than options?

	Futures	Options
Exposure	Linear —profit and loss move one-for-one with the underlying	Non-linear; depends on strike and moneyness
Extra drivers	None	Implied volatility, time decay, strike
What you are betting on	Direction	Direction and volatility

An option position is partly a volatility trade whether you intended it or not. Linear exposure is what lets one risk framework size 37 positions on the same scale.

Example 1.8. A trend follower might simultaneously hold long ES, short ZN, long GC, short 6J—all four are futures, and all four are different asset classes.

Appendix · Notation

Throughout, t is the date and i indexes an index’s constituents.

Symbol	Means	First used
I_t	the index level on date t —a published number, not a tradeable object	§1.1.1
P_{it}, N_i	constituent i ’s price on date t , and its float-adjusted share count	§1.1.1
M_t, D_t	the total float capitalisation, and the divisor that turns it into the printed level	§1.1.1
w_{it}	constituent i ’s share of total capitalisation—its weight in the index return	§1.1.1

Symbol	Means	First used
r_I, r_i	the index's return and constituent i 's return, over the same period	§1.1.1
s	the signed share count of a position: positive is long, negative is short	§1.3.3
P_0, P_T	the entry and exit price of that position	§1.3.3
$\text{PnL}(P_T)$	the position's profit and loss as a function of where the price ends up	§1.3.3
N	the number of shares borrowed to open a short	§1.3.2

Note (Collisions to watch). s here is a **signed share count**, not 2's s , which indexes the asset. w_{it} is a **constituent's weight inside an index**, fixed by capitalisation and not chosen by anyone; 2's $w_{s,t}$ and 5 · §3.3's w_i are **portfolio** weights, which are decisions. r_i is a constituent's return, whereas elsewhere in the course r carries an asset and a date subscript and means the forward return being predicted.

2 · Testing a Signal

2.1 A signal is a hypothesis, not a formula

2.1.1 The three objects

Everything in this chapter is built from three numbers, one of each per asset per date.

Object	Symbol	What it is	Known at t ?
Weight	$w_{s,t}$	the share of capital held in asset s on date t , signed—negative is a short	yes , you choose it
Signal	$MOM_{s,t}$	a number computed from data available at t , meant to say something about what comes next	yes , you compute it
Forward return	$r_{s,t}$	what the asset then goes on to deliver while the position is held	no —this is the unknown

Note (Scope). This chapter describes **one asset at a time**. That is not a simplification: assets have different volatilities, so their raw signals are not on one scale and cannot be compared until §2.4 puts them there—and combining several into one book comes after that, in the position-construction step.

Definition 2.1 (Binary momentum). The simplest member of the family—the sign of last period’s return, and nothing else:

$$MOM_{s,t} = \text{sign}(r_{s,t-1})$$

where s indexes the asset and t the date, $r_{s,t-1}$ is that asset’s return over the period just ended, and $MOM_{s,t}$ is either +1 (long) or −1 (short), with nothing in between.

Trend direction survives; trend strength is thrown away. A window in which the asset rose 20 percent and one in which it rose 10 percent produce the *same* signal, so a strategy built on it takes the same size in both.

Definition 2.2 (Momentum). Averaging over N periods rather than one, and keeping the value rather than its sign:

$$MOM_{s,t} = \text{Avg}(r_{s,t-i}) = \frac{1}{N} \sum_{i=1}^N r_{s,t-i}, \quad i = 1 \dots N$$

with N the lookback length and i the lag inside it—the average runs over the N returns ending yesterday, so at $N = 21$ it is last month’s average daily move.

Example 2.3. Three assets over a five-day lookback:

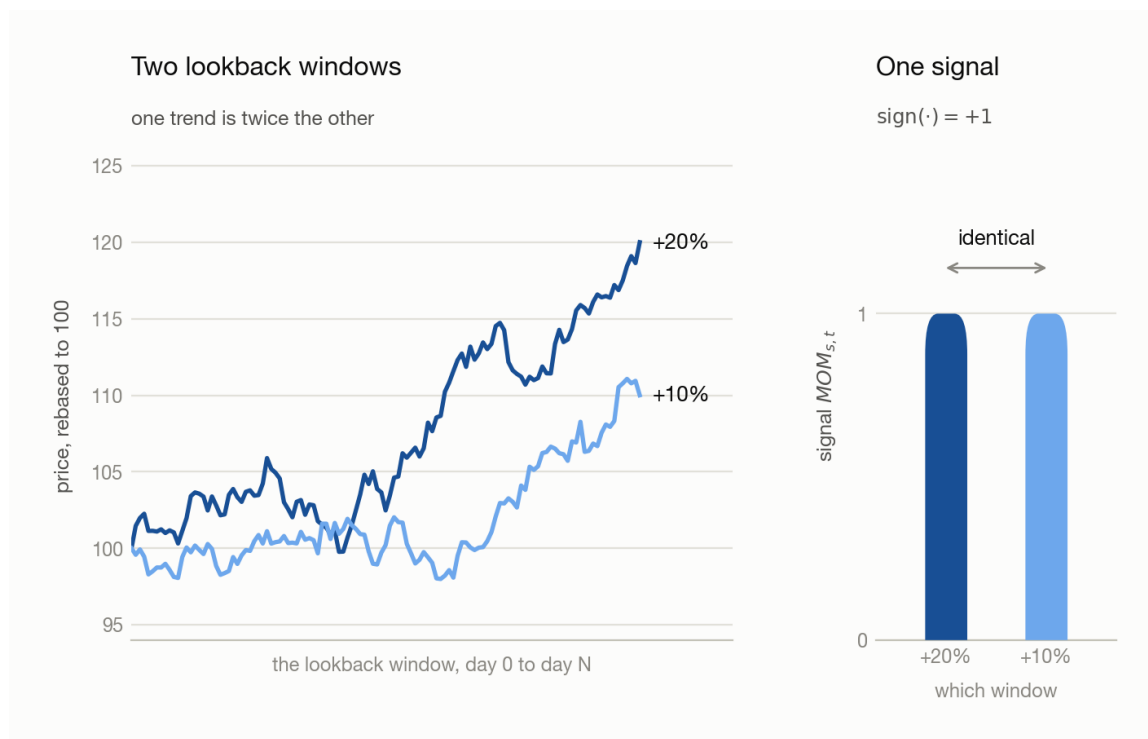


Figure 2.1: Left, one asset’s price over two lookback windows, rebased to 100: one climbs to 120 and the other to 110. Right, the signal each path produces—two bars of identical height at +1. The magnitude of the move does not reach the signal. Illustrative.

Asset	Last five returns	Avg → the signal	After sign
A	+3%, +2%, +4%, +1%, +2%	+0.024	+1
B	+1%, −0.5%, +1%, 0%, +0.5%	+0.004	+1
C	−2%, −3%, −1%, −2%, −2%	−0.020	−1

The right column cannot tell A from B; the left makes A **six times** B, and since §2.1.2 sets each weight proportional to its signal, A is held six times larger. That ratio is the magnitude, and it is all the dropped sign was costing.

Note (The level is not yet meaningful). 0.024 means nothing on its own—2.4 percent in a calm year and 2.4 percent in a violent one are not the same trend. Only *relative* sizes are ever used, and §2.4 is what puts them on one scale.

2.1.2 Why studying the signal is the same as studying the portfolio

A signal is not a strategy: it earns nothing and cannot be held. The reason to spend a chapter on one runs backwards, from what you are actually paid on—**so start from the portfolio, because that is what you optimize.**

$$R_t = \sum_s w_{s,t} r_{s,t}$$

R_t is the portfolio's return on date t , summed over every asset in the universe. Of the two factors in each term only one is yours— $r_{s,t}$ is the market's answer, and it arrives after the weight is already set. **So the whole of portfolio construction is the choice of $w_{s,t}$.**

But the weights are not free. Two sums over the book are fixed before any signal is consulted—the **net**, fixing how much *market* the book carries since longs and shorts cancel in it, and the **gross**, fixing how much *leverage* is deployed since they add:

$$\text{net: } \sum_s w_{s,t} = 1 \qquad \text{gross: } \sum_s |w_{s,t}| = G$$

with G the gross target—200 percent on a 150/50 book—and the net holding at 1 only to within a tolerance δ , since rebalancing is discrete. The two are independent: one asset at 100 percent and a book long 200 percent / short 100 percent carry the same net and three times the position, which is why gross is never free (1).

The weights are where the signal enters. You want weight where the forward return is about to be high, and that is precisely the number you do not have. The signal is the stand-in you put in its place, which means setting $w_{s,t} \propto MOM_{s,t}$.

Claim 2.4. Under that rule the portfolio's expected return is proportional to the correlation between the signal and the forward return, and to nothing else that can change its sign.

Proof. Take the signal and the return as centred. With $w_{s,t} \propto MOM_{s,t}$,

$$E[R_t] \propto \sum_s E[MOM_{s,t} r_{s,t}] = \sum_s \rho_s \sigma_{MOM,s} \sigma_{r,s}$$

using $E[xy] = \rho \sigma_x \sigma_y$ for centred x and y , where ρ_s is the correlation between asset s 's signal and its forward return and $\sigma_{MOM,s}$, $\sigma_{r,s}$ their standard deviations. The dropped constant is a leverage choice and the volatilities are measurable properties of the asset—all three positive whatever the signal does. **Only ρ_s can carry a sign, so only ρ_s decides whether the portfolio makes money or loses it.** $\square$

Note (The constraints do not disturb this). Reaching a legal weight vector means **shifting** the signal until the net lands right and **scaling** until the gross does—both affine with a positive scale, under which correlation is unchanged: $\rho(a + bx, y) = \rho(x, y)$ for $b > 0$. The ρ measured on the raw signal is therefore the ρ governing the constrained portfolio, which is what lets the rest of the chapter ignore weights altogether; the position-construction step carries out both operations.

Everything collapses to that one number, and it is a number about the signal alone:

$$\text{hypothesis: } MOM \uparrow \implies \text{return} \uparrow \qquad \text{that is: } \rho > 0$$

Write that down before the code—it names what would falsify the signal, and one you cannot falsify is a plot you will rationalize either way. Two things follow from it, and between them they set the shape of the rest of the chapter:

Consequence	Dealt with in
The portfolio never has to be built to test the signal. Measure ρ between the signal and the forward return directly, and you have measured the strategy	§2.2, §2.3
At equal ρ a volatile asset contributes more than a quiet one , since each term carries $\sigma_{MOM,s}\sigma_{r,s}$ alongside ρ_s . Ranking raw signals therefore ranks volatilities	§2.4

2.2 What you hoped for, and what you get

2.2.1 Which return the signal is scored on

Before signal and return can be plotted against each other, *the forward return* has to be pinned down: §2.1.2 measures ρ against it, but the phrase is not yet precise enough to compute. Three spans sit on the timeline, in this order:

Span	What it holds	Typical size
Lookback	the N returns the signal averages, ending at $t - 1$	20–250 periods
Gap	g periods thrown away—neither averaged into the signal nor scored against it	0, a week, or a month
Paired return	$r_{s,t+g}$, the one period the signal is judged on	one period

The gap is the only free choice of the three—the -1 in **12-1 momentum** is exactly this gap, a twelve-month window stopping one month short of today.

Why leave one at all. Not executability: §2.1.1’s window already ends at $t - 1$, so $g = 0$ is knowable in time and anything tighter is look-ahead (§2.6). Executability fixes the floor at zero; **reversal** decides everything above it. A trend that has just formed hands a little of it back, and at daily frequency that rebate is large enough to cancel the edge outright—which is how a real signal arrives at the bar plot looking flat, or upside down. The mechanisms behind it are this chapter’s Background.

Both directions cost something, which is what makes g a **hyperparameter** rather than a constant. Reversal strengthens as the sampling gets finer—the bid–ask bounce is a roughly fixed number of ticks per trade while the trend grows with the horizon—but a wider gap also throws away the opening of the trend. Set it from your trading frequency and a prior on how long the rebate lasts; running it at a day, a week and a month and keeping whichever bar plot looks best is what 5 · Modelling warns about.

Note (Neighbouring observations are near-copies). Step one date forward and the lookback keeps all but one of its cells, as the figure’s lower panel shows. Consecutive

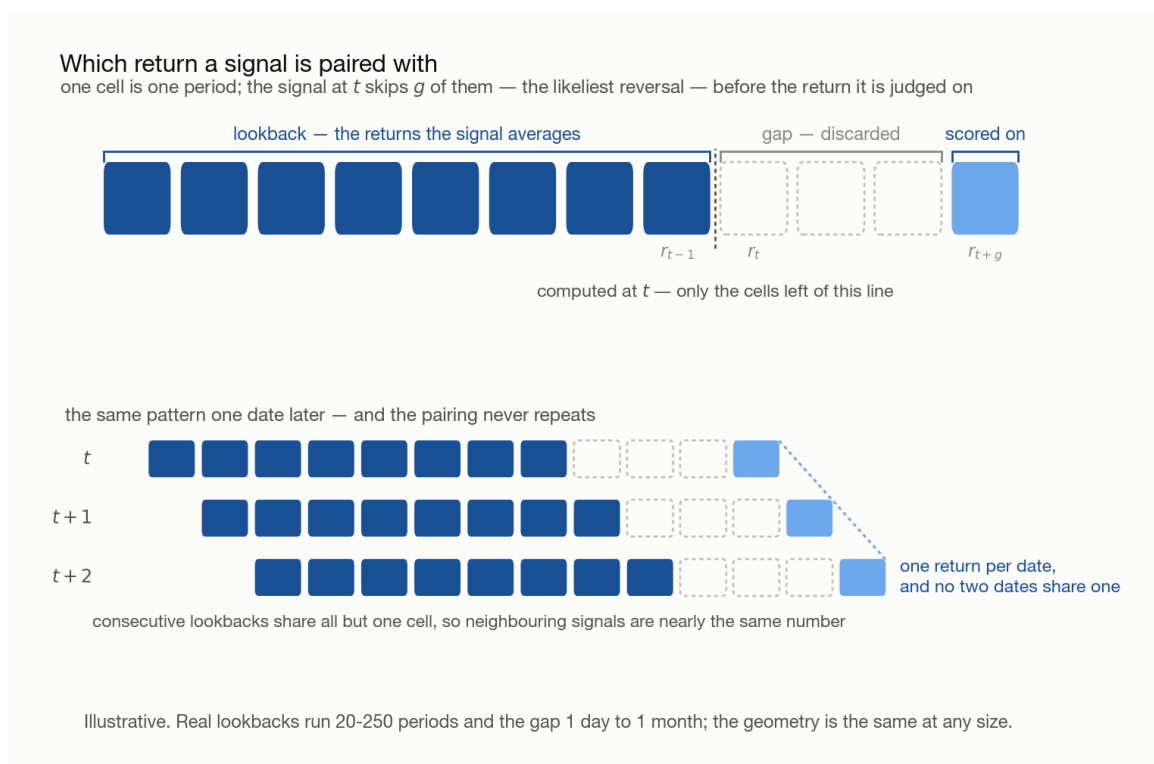


Figure 2.2: Above, one observation laid on a timeline of one cell per period: the lookback the signal averages, then the discarded gap g , then the single period the signal is scored on. Below, the same pattern for dates t , $t+1$ and $t+2$, each shifted one cell right—the lookbacks overlap in all but one cell, while the three scored cells form a descending diagonal, so no two dates share a scored return. Illustrative.

signals are therefore not independent draws, which is where §2.3.2.2's caveat on the effective sample size comes from.

2.2.2 The plot everyone draws first

§2.1's hypothesis says forward return rises with the signal, so the first move is to plot one against the other. The picture you had in mind is the leftmost panel below. What comes back—for well over 99 percent of the scatters you will ever draw—is the rightmost.

2.2.3 Why the real one is a cloud

Claim 2.5. The scatter can neither confirm nor refute a signal, because the correlation a working signal carries sits far below the threshold at which the eye resolves a trend.

Proof. Both numbers are readable off the panels above: the eye stops resolving a tilt somewhere around 30 percent, and the fourth panel—21-day risk-adjusted momentum

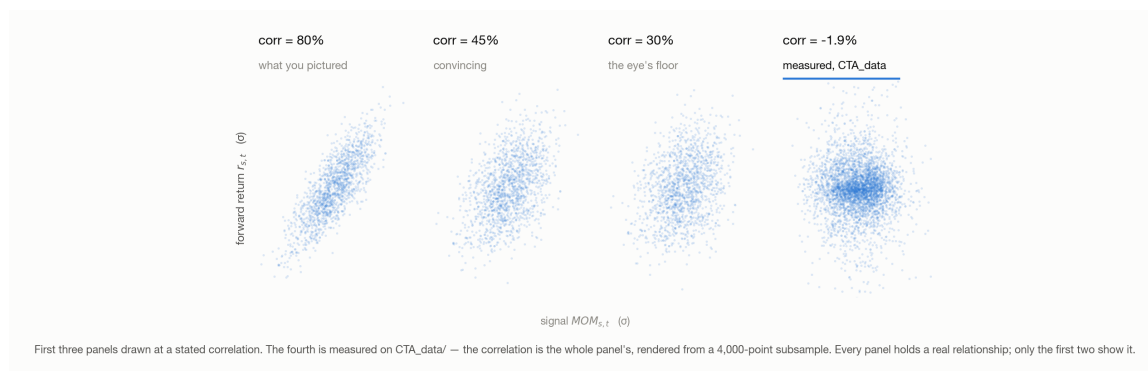


Figure 2.3: Four scatters of forward return against a signal, both axes in standard deviations. The first three are drawn at correlations of 80, 45 and 30 percent from synthetic points; by 30 percent the tilt is barely arguable, which is the eye’s floor. The fourth is measured on `CTA_data` at -1.9 percent—a formless cloud with a denser core, being real returns with fatter tails.

against the next session’s return, measured over the 37 ETFs of 100 · The Dataset—comes in at **-1.9 percent**. Since $1.9\% < 30\%$, a signal that works and a signal that does not produce the same picture—the **absence of a visible trend is not evidence of anything**. $\square$

So why is the correlation so small? Because the quantity being plotted is mostly not the signal.

Claim 2.6. Split the forward return into the part the signal reaches and the part it does not, $y = \beta x + \epsilon$. If ϵ is uncorrelated with x , the signal owns exactly ρ^2 of the return’s variance, where ρ is their correlation.

Proof. Uncorrelated components add their variances,

$$\text{Var}(y) = \beta^2 \text{Var}(x) + \text{Var}(\epsilon)$$

and on one regressor with an intercept the explained share is $R^2 = \rho^2$. At $\rho = -0.019$, $R^2 = 0.00035$ —which does not mean “right 0.035 percent of the time”, but: of the variation in forward return from one observation to the next, **0.035 percent is linear in the signal and 99.965 percent is not**. $\square$

Example 2.7. Everything below is **one asset-date’s** forward return, never a portfolio’s, and every number is measured over the 57,649 asset-dates of the sample. The daily return standard deviation is $\sigma_y = 0.0149$ —149 bp, a **basis point** being 0.01 percent:

Component	Size	At $\rho = -1.9\%$
Total return	σ_y	149 bp
What the signal moves	$\rho\sigma_y$	2.8 bp
What it does not	$\sigma_y(1 - \rho^2)^{1/2}$	149 bp

The third row is not a rounding slip: at this correlation the signal removes so little that the unreachable part is the whole of it to three figures. Across an x -axis running from $-2\sigma_x$ to $+2\sigma_x$ the trend line moves about 11 bp end to end, while at every x the points scatter over roughly 596 bp. **A 0.11 percent slope inside a 6 percent cloud**—that ratio *is* the picture.

Note (Where a larger correlation would come from, and where it would not). A predictor correlating 50 percent with next period's return would be arbitraged away long before you found it in anything liquid, so single digits is a competitive market's ceiling rather than a shortfall in craft. Two things do legitimately raise it and neither is on offer here: a **longer horizon**, since the signal's edge accumulates while the noise grows only as the square root of time, and **combining many signals**, since a composite reaches correlations no ingredient has on its own. Quoted numbers near 10 percent are almost always one of those two. A single signal against a single session is the hardest case in the family, and this chapter is deliberately standing in it.

2.3 Two ways out of the cloud

The scatter is unreadable, but the signal may still be in it. There are two things to try, and only the second one works.

2.3.1 Turn down alpha

alpha is a marker's opacity. Turn it far enough down—`alpha=0.05` on a few thousand points, lower still on more—and a single point becomes nearly invisible, so only *overlap* renders and the chart is a density map rather than a mass of ink.

Note (Not the other alphas). matplotlib's opacity keyword: not a regression intercept, not the excess return a manager is paid for, not 3's smoothing constant α . Code font is the tell.

Sometimes it is enough. A corner bitten out of the cloud—high signal never followed by a bad loss—would be a finding, because a signal that only rules something *out* is still tradeable. This data has no such corner, and that is the ordinary outcome.

Why it usually fails. Two reasons that compound:

- **The density comes back smooth.** Tens of thousands of asset-date cells average into a clean bivariate blob, and opacity renders that faithfully—a smooth density has no feature, and §2.2's band is fifty times taller than the tilt hiding in it.
- **It treats the symptom.** The scatter spends its resolution on individual noisy points when the claim is about their **average**; no rendering choice changes what is being rendered.

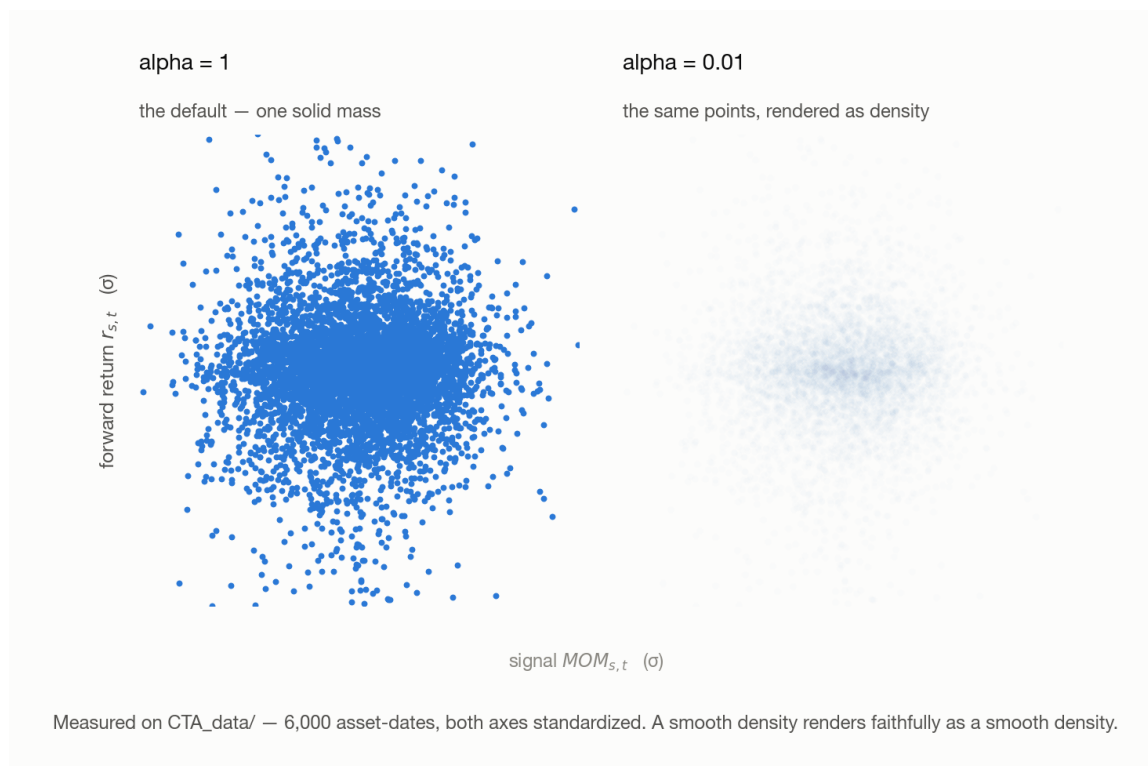


Figure 2.4: The same measured asset-dates twice, both axes standardized. Left, at full opacity, they render as one solid disc with no internal structure. Right, the identical points at an opacity of 0.01: a smooth round density, denser in the core, with no thin region, no empty corner and no visible tilt anywhere.

Note (Look first anyway). Always draw it. On the rare occasion something *is* readable—a curve, an empty corner—it beats every statistic, because it gives the *shape*. The mistake is concluding anything **from** a cloud.

2.3.2 Beta Method

Stop asking each point what it did, and ask each *group* of points what they did on average. The answer comes back as a **bar plot**, and the rest of §2.3 is about reading that one chart.

2.3.2.1 What a bar plot is

Definition 2.8 (Bar plot). Five bars. The horizontal axis is the **signal**, coarsened into five ordered slots G1 ... G5; the vertical axis is the **mean forward return** of the observations filed into each slot. Every observation lands in exactly one bar, and the error bar on it is the standard error of that mean.

Note (A bucket holds dates, not assets). G5 is not a set of assets. It holds the **dates** on which this one asset's signal stood in the top fifth of its own history, and G1 the dates on which it stood in the bottom fifth. One asset passes through every bucket as the years go by, and the whole chart describes that asset alone.

Five steps build it:

1. **Fix one asset.** Its history gives one signal $MOM_{s,t}$ and one forward return $r_{s,t}$ per date.
2. **Rank the dates by signal value**, highest to lowest.
3. **File each date** into one of five slots by that rank: the highest fifth into G5, down to G1.
4. **Record** under each slot the forward return that date went on to deliver.
5. **Average** the returns filed under each slot.

Note (Step 2 is where this chapter’s one defect lives). Ranked on *what*, exactly? Take the raw momentum value and everything above still runs—but the asset’s own volatility changes over the years, so +2 percent is a strong month in a calm tape and a quiet one in a violent tape, and step 3 files them into the same slot as if they were the same event. §2.4 is the correction, and until it lands the plot below should be read as machinery rather than as a result.

Note (What makes 57,649 dates from 37 assets one pool). The measured row does not rank raw momentum across assets. Every signal entering it is already divided by that asset’s own trailing volatility and ranked against that asset’s own history before it lands in a bucket—the route §2.4 works out in full, with a worked two-asset example. G1 ... G5 are therefore **percentile** slots, comparable across a Treasury ETF and a semiconductor ETF for the same reason they are comparable across 2021 and 2023: nothing but rank survives the trip, so pooling assets costs no more than pooling dates already did.

The sort key is the signal, never the return. Step 2 orders by $MOM_{s,t}$, which is known at t ; step 3 only *records* what followed. G1 therefore holds the lowest-**signal** observations, not the worst performers.

Example 2.9. Five dates from one asset’s history, invented, filed both ways:

Date	Signal	Forward return	Slot by signal	Slot by return
t_1	+2.1%	−0.4%	G4	G2
t_2	−1.8%	+0.9%	G1	G4
t_3	+0.6%	+1.3%	G3	G5
t_4	+3.4%	+0.2%	G5	G3
t_5	−0.9%	−1.1%	G2	G1

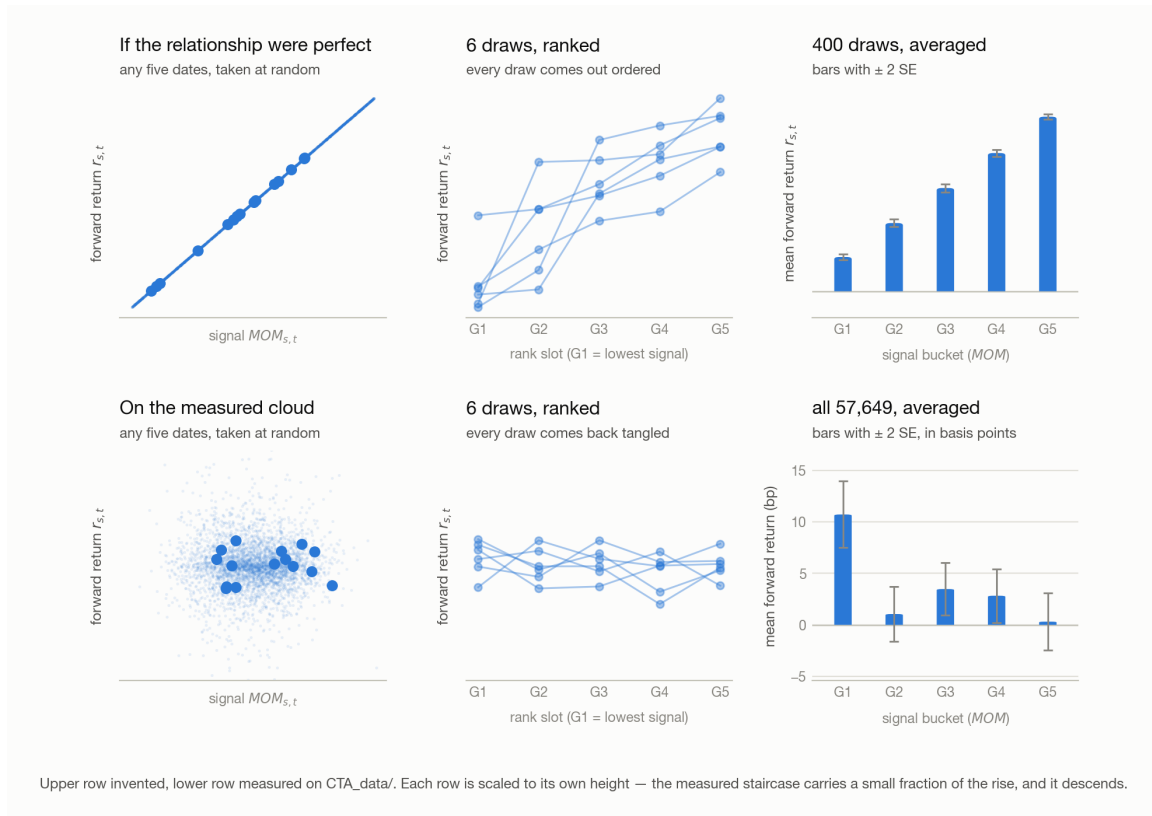


Figure 2.5: Six panels; the upper row is illustrative and the lower is measured on CTA_data. Left, the population of dates with fifteen drawn at random marked: on a perfect relationship they sit along a line, on the measured cloud anywhere. Middle, six draws of five dates plotted against the slots G1 to G5: on the line every draw rises monotonically, on the cloud the six tangle. Right, the answer with ± 2 standard errors—a clean rising staircase above, and below the real chart, where G1 stands well clear of its error bar and G2 to G5 sit within a few basis points of zero, so the staircase descends.

Read down *slot by signal* and the returns come out unordered—which is exactly what five dates look like at a correlation this small. Read down *slot by return* and the ordering is perfect, and would be perfect for **any** signal including one straight out of a random number generator, because each bar is then reporting the sort key back to you. **The staircase is evidence only because the thing sorted on and the thing measured are different, and the second was not knowable when the first was computed.**

One bar therefore carries three pieces of information: which fifth of the signal's range it stands for, the average return that fifth went on to earn, and how much of that average is sampling noise.

2.3.2.2 How the bar plot shows a trend

The trend is the ordering, not the heights. §2.1's hypothesis was *higher signal, higher forward return*; on a bar plot that is precisely the statement $G1 < G2 < G3 <$

$G4 < G5$. A single bar's level moves with where the boundaries are cut (§2.4)—the ordering is what does not.

Two further readings carry information, and nothing else on the chart does:

Read	What it establishes
G5 – G1, measured against the error bars	the size of the edge next to what noise alone would draw. A rise that fits inside one error bar is not evidence of anything, and §2.3.2.3 turns that comparison into a test
The direction of the slope	a <i>descending</i> staircase is not a dead signal—it is the same edge carrying the opposite sign

A scrambled middle does not disqualify a signal: clean tails around a muddled G2–G4 is a common shape and a perfectly tradeable one, since the tails are where the positions go.

On direction, adopt a convention and keep it: **leave the signal's sign alone and let the chart come out upside down.** Flipping a signal the moment its staircase descends is cheap, and a dozen signals later you will not remember which ones you flipped. Read a descending staircase as *this works, traded short*, and carry the sign in the strategy rather than in the signal definition.

Claim 2.10. The ordering surfaces at all only because of step 4—averaging shrinks the noise and leaves the signal untouched.

Proof. Take m observations sharing a similar signal value. Their mean return still has expectation β times their mean signal—they were chosen for having nearly the same signal, so averaging them changes it barely at all—while the noise around it falls to $\sigma_\epsilon/m^{1/2}$. Measured, with the edge taken as the G5 – G1 gap the chart is trying to resolve:

	One observation	Mean of 11,522
Signal	10.4 bp	10.4 bp
Noise	149 bp	$149/11522^{1/2} \approx 1.4$ bp
Ratio	1 : 14	7 : 1

The left column is the scatter of §2.2 and the right column is one bar of the chart. Nothing was added—only the noise was taken away. $\square$

The quantity being estimated is identical in all four panels—the real bucket means, +11 bp at G1 and roughly zero at G5. Only the error moves, and at $m = 30$ it is five times the whole staircase. **Sample size is not a detail of the recipe; it is the reason the recipe works.**

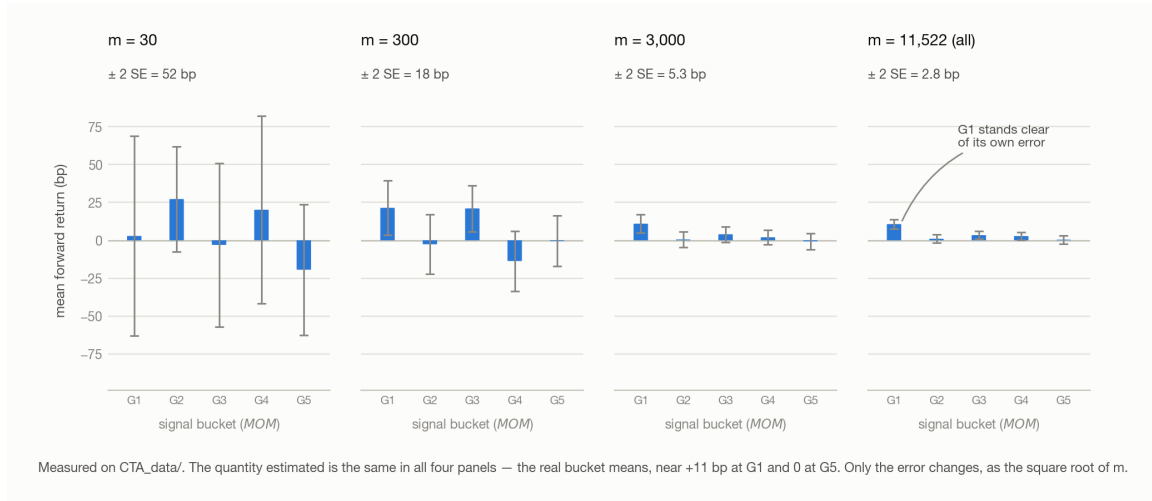


Figure 2.6: The same measured population as four bucket charts, mean forward return in basis points against signal bucket, drawn from 30, 300, 3,000 and all 11,522 observations per bucket. At $m = 30$ the bars swing over tens of basis points with error bars wider still and no ordering; by $m = 3,000$ G1 is visibly the tallest bar; at $m = 11,522$ the error bars are a few basis points and G1 stands clear of its own error while G2 to G5 sit near zero.

Note (Where that overstates it). The $m^{1/2}$ assumes independence, and one asset's history violates it twice over: overlapping lookback windows share most of their inputs, and returns cluster in time, so consecutive observations are near-copies rather than independent draws. Both push the effective count well below m , and the noise does not really reach 1.4 bp. The logic survives: averaging cancels noise and leaves systematic signal.

2.3.2.3 Is the staircase real?

§2.3.2.2 read the rise against the error bars by eye. That comparison is a hypothesis test, and it is worth writing down, because the eyeball version and the arithmetic version do not agree.

Definition 2.11 (Standard error of a bucket). For bucket g holding m_g observations whose forward returns have sample standard deviation σ_g ,

$$SE_g = \frac{\sigma_g}{m_g^{1/2}}$$

the spread of that bucket's **mean**, not of the observations inside it—which is why it shrinks with m_g while σ_g does not. Bars here are drawn at $\pm 2 SE_g$.

The hypothesis on trial is §2.1's, narrowed to the two buckets that carry the positions. A long book buys G5 and sells G1, so what has to be non-zero is the difference between those two means:

$$H_0 : \mu_{G5} = \mu_{G1}$$

against the alternative that they differ, where μ_g is bucket g 's true mean forward return. The statistic is the observed difference in units of its own noise:

$$z = \frac{\mu_{G5} - \mu_{G1}}{(\text{SE}_{G5}^2 + \text{SE}_{G1}^2)^{1/2}}$$

the denominator adding **variances** rather than standard errors, because the two bucket means are computed from disjoint sets of dates.

Note (Why it is a t and gets called a z). Both μ_g and σ_g are estimated from the same sample, so the exact null distribution is Student's t , not the normal. It stops mattering fast: at bucket sizes in the thousands the two are indistinguishable and the number is quoted as a **z-score**. Derive it as a t , report it as a z , and at m_g in the tens use the t and mean it.

Claim 2.12. $|z| \geq 2$ rejects H_0 at the 5 percent level.

Proof. Under H_0 the statistic is standard normal in the large-sample limit, and a standard normal places 0.0455 of its mass outside ± 2 —less than $\alpha = 0.05$, so the observation lies inside the two-sided rejection region. $\square$

Example 2.13. The whole sample, five buckets, 11,522 dates in the smallest:

Bucket	μ_g	σ_g	m_g	SE_g
G1	+10.72 bp	173 bp	11,522	1.62 bp
G2	+1.05 bp	144 bp	11,523	1.34 bp
G3	+3.48 bp	136 bp	11,525	1.27 bp
G4	+2.83 bp	140 bp	11,523	1.30 bp
G5	+0.31 bp	149 bp	11,556	1.39 bp

$$z = \frac{0.31 - 10.72}{(1.39^2 + 1.62^2)^{1/2}} = \frac{-10.41}{2.13} \approx -4.9$$

Well past 2 in absolute value, and the bars—G1 spanning +7.5 to +14.0 bp, G5 spanning −2.5 to +3.1—clear each other with room to spare.

Note (Non-overlap is sufficient, not necessary). The two readings differ, and in the direction that costs you signals. Bars of half-width 2 SE fail to touch when the difference exceeds $2(\text{SE}_{G5} + \text{SE}_{G1})$, while the test rejects when it exceeds $2(\text{SE}_{G5}^2 + \text{SE}_{G1}^2)^{1/2}$ —and a sum is never smaller than the root of the sum of squares. On the measured numbers the eye demands 6.0 bp and the test demands 4.3 bp, so a difference between those two is one the picture calls inconclusive and the arithmetic calls significant.

Bars that clear each other are always significant; bars that touch may still be. Use the chart to accept and compute z before rejecting.

The sign is negative, and that is a result rather than a failure. $z = -4.9$ says the two means differ, decisively; it says nothing about which way round they should be. What was measured is that **a high 21-day momentum is followed by a slightly worse next session than a low one**—the descending staircase §2.3.2.2 told you to expect and to keep. It is not a broken signal, it is **reversal**, and the Background section explains where it comes from and what removes it: put a gap between the signal’s window and the return it is scored on, and the sign comes back.

Note (What the test does not license). Two things sit outside it. The $m_g^{1/2}$ in every standard error assumes independent observations, and §2.3.2.2’s closing note explains why one asset’s overlapping windows are nothing of the kind—the effective count is well below m_g , so every z here is optimistic. And a threshold of 5 percent means one signal in twenty passes on noise, which is a fact about *one* test: running twenty variants and reporting the one that cleared 2 is not a test at all, and is 5 · Modelling’s subject.

Failing to reject is a verdict on the measurement, not on the signal. $|z| < 2$ says the data did not separate the two means. Whether that is because they are equal, or because the error bars were too wide to tell them apart, is a question about m_g —and the two readings are not interchangeable. The smallest gap the test could have caught, at the four sample sizes of the figure above and a common σ_g of 149 bp:

m_g	SE_g	Smallest detectable G5 – G1	What a null result there rules out
30	27 bp	77 bp	Almost nothing. An edge large enough to build a career on is still consistent with it
300	8.6 bp	24 bp	Only edges several times larger than anything this chapter measures
3,000	2.7 bp	7.7 bp	Most of the range that matters, but not the part costs decide
11,522	1.4 bp	3.9 bp	The effect, down to a gap too small to survive execution anyway

Only the bottom row turns a null result into a finding. In the top row the gate has not been failed so much as attempted—the **honest report is “not measured yet”, and the fix is more observations, not a different signal.**

2.3.2.4 What the bar plot cannot show

Every date contributes one observation—its signal’s score, and the return that followed. Step 5 flattens all of them onto one picture, five slots wide.

Step 5 integrates over t , and an integral hands back an area, never the shape of the function underneath it. A staircase that is monotone over twenty years is equally consistent with an edge that held throughout and with one that worked for five years and was flat for fifteen.

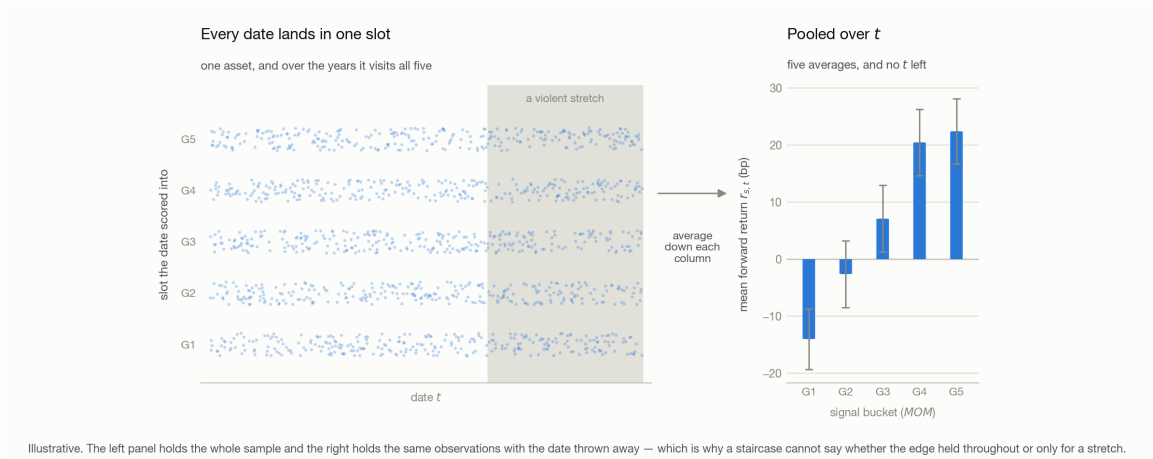


Figure 2.7: Left, one asset’s dates strung along a time axis, each drawn at the slot its signal scored into, so all five rows are populated throughout and a violent stretch at the right is shaded. Right, the same observations with the date discarded: five bars of mean forward return with error bars. The arrow between them averages down each column, and the time axis is what it spends.

Not on the chart	Where to look instead
When the edge happened	the equity curve of 5 · Modelling
Who is inside a bucket—which dates, which regime	the dates behind each bar, tabulated rather than averaged
The spread of returns behind a bar, as against its mean	the distribution within a bucket, not its mean
How independent the observations are	overlapping windows, per the note above

2.4 Step 2, corrected — risk-adjusted momentum

This section repairs §2.3.2, it does not extend it. Every piece of machinery there survives: the sort key is knowable at t , the returns are recorded honestly, and the averaging shrinks noise exactly as the proof says. What §2.3.2 never settled is *what step 2 ranks*, and ranking the raw momentum value produces a chart about something other than the signal.

Raw momentum is not comparable **across time**, and §2.3.2 compares nothing else: its five slots sort one asset’s own dates against one another. 2 percent in the calm 2021 tape and 2 percent in the 2023 rate-hike drawdown are different events, and filing them into the same slot says they are the same.

The identical failure reappears one step later, **across assets**, the moment you want more than one name in a book—and there it is an assumption you have to state rather than a fact. A universe of five hundred large-cap equities might plausibly sit

on one scale already. A CTA universe does not: 50 bp in a session is an ordinary day for an equity index and, for a Treasury future, an event that would lead the news. 2 percent monthly momentum in a Treasury ETF is a large move; the same 2 percent in a semiconductor ETF is a quiet month. Rank the two against each other on raw momentum and the semiconductor wins every time—not because its trend is stronger but because everything about it is larger.

Divide by volatility:

$$MOM_{s,t}^{\text{risk-adj}} = \text{Avg} \left(\frac{r_{s,t-i}}{\sigma_{s,t}} \right), \quad i = 1 \dots N$$

where $\sigma_{s,t}$ is that asset’s volatility, estimated on data strictly before t —a denominator that peeks at the future contaminates the signal as surely as a numerator would (§2.6).

Every date, and later every asset, now lands on one scale, so §2.3.2’s slots compare like with like—two students both scoring 80 on different exams against different cohorts have not achieved the same thing.

Note (A second route to the same plane). Dividing by $\sigma_{s,t}$ is not the only way. You can also replace the value outright by **its percentile against that asset’s own past**, which lands every date on $[0, 1]$ by construction and needs no volatility estimate at all. Either route hands step 2 something it can legitimately rank.

Example 2.14. Two assets, each with five past signal values and one for today:

Asset	Its own past signals	Today	Beats	Percentile
A	−10, 8, 4, 0, −5	+7	four of five	80
B	−100, −80, 90, −60, 20	−70	two of five	40

Raw, +7 against −70 is not a comparison anybody should make. As percentiles, 80 against 40 is—and it says A is the stronger trend *relative to its own history*, which is the only sense in which the question has an answer. The cost is the one every rank charges: the magnitudes are gone.

Note (Which cut to use once it is scored). The percentile route hands back a uniform score, so cutting at the quintiles gives five equal groups for free. The volatility route does not: a standardized signal is roughly normal, so cutting it at fixed intervals such as ± 1 and ± 2 leaves almost everything in the middle and starves the two tails you would actually trade. Cut it at its quintiles instead.

The rule this section leaves behind. Every signal is **standardized** or **scored against its own past** before it is compared to anything—otherwise the bar plot reports volatility while wearing the signal’s name. The measured chart back in §2.3.2 already followed this rule before this section stated it: the percentile route is what let 37 assets share one pool of buckets.

2.5 What to report — and why it is not a Sharpe ratio

Every chart in §2.3 reports one quantity: **the mean forward return of a bucket, with its standard error**. The instinct at this point is to reach for something that sounds more like a result, and the one everybody reaches for is the wrong tool for this stage.

Definition 2.15 (Sharpe ratio). A portfolio’s average return in excess of what financing it costs, per unit of the volatility it ran:

$$\text{SR} = \frac{E[R_p] - r_f}{\sigma_p}$$

with R_p the portfolio’s return over a period, r_f the risk-free rate charged on the capital it used, and σ_p the standard deviation of R_p .

It is the right measurement for a finished portfolio and the wrong one here, for four separate reasons.

There is no portfolio yet. §2.1.2’s argument is precisely that the signal can be judged *without* building one, and nothing in §2.3 has chosen a weight. A Sharpe ratio scores the output of a construction step that has not happened; computing one on a bucket’s returns scores a portfolio nobody would run—equal weight on every date that fell into one fifth of the sample.

The financing rate is not zero here, and it is not constant. The numerator subtracts r_f , and which rate that is depends on the book’s **net exposure** (§2.1.2):

Book	Net exposure	Correct r_f
Dollar-neutral long/short—one dollar long against one dollar short	≈ 0	zero. The short leg’s proceeds fund the long leg, so no capital is being borrowed
A 150/50 CTA book	100 percent	the actual financing rate , on the capital the net exposure represents

Statistical-arbitrage books are the first row, which is where the habit of setting $r_f = 0$ comes from and where it is correct. A CTA book is the second row, and the rate belonging in it is a **time series**, not a constant: the policy rate this sample spans moved from near zero to above five percent. Carrying over “we always used 2 percent” from a book that was dollar-neutral puts an invented series into the numerator of every number reported.

It folds three estimates into one number. Mean, volatility and financing rate, each with its own error. A Sharpe that falls because the edge died and a Sharpe that

falls because the tape got noisier are the same number—the exact opposite of §2.3.2’s method, where one chart carries one quantity so a change has one cause.

A Sharpe means nothing without the market it was earned in. Two books, and the smaller number is the better strategy:

The tape	Market’s own Sharpe	The book’s Sharpe	Reading
Violent throughout	0.4	0.8	held together through amplitude that should have wrecked it, at twice the market’s own risk-adjusted return—an excellent result
Calm and rising	0.8	0.9	barely beat holding the index. The extra 0.1 is within what one lucky bet would produce

The mean return is the quantity you can actually trade. It is what the book delivers, it is what §2.1.2’s algebra is written in, and it is denominated in the same basis points as the costs that will be subtracted from it later. A Sharpe ratio is a score attached afterwards, useful for comparing **finished** portfolios against each other—which belongs to performance evaluation, after an optimization step has given it something to measure.

2.6 Information availability

Everything above assumes the signal at t uses only data knowable at t . Look-ahead bias is born here; the execution offsets of 5 · Modelling are the second line of defense and cannot rescue a signal contaminated at construction. The rolling-quantile rule (§2.4) and the train/validation/test split (5 · §1) are the same discipline.

Background

Why does a trend hand part of itself back, and can that be competed away?

Reversal is not an anomaly, it is the normal state of the tape. A trend that has just formed hands a little of it back, and at daily frequency that rebate is large enough to cancel the edge outright—which is why §2.2.1 leaves a gap g between the signal’s window and the return it is scored on. Three mechanisms produce it, and they are not equally fragile:

Mechanism	What happens	Competed away?
Bid–ask bounce	Closes print alternately at the bid and the ask, oscillating around the true mid—negative serial correlation by construction	No. A measurement artefact; no price moved
Paying for liquidity	An order that must fill now pushes price past fair value, and whoever takes the other side is repaid when it returns	No. A fee for a service actually rendered
Overreaction	News is over-extrapolated, then partly corrected	Yes —the only part that decays

Two of the three are not mistakes, which is why reversal survives competition instead of being arbitrated into nothing—only the overreaction row decays as more people trade against it.

Appendix · Notation

Throughout, s indexes the asset and t the date. The signal is computed per asset, so s is dropped wherever a formula never leaves one asset; it carries weight everywhere the assets are ranked against each other.

Symbol	Means	First used
s, t	the asset, and the date in periods (days here)	§2.1
$r_{s,t}, MOM_{s,t}, w_{s,t}$	that asset’s return in that period, the momentum signal it produces, and the share of capital that signal earns it	§2.1.1
R_t	the portfolio’s return on that date, $\sum_s w_{s,t} r_{s,t}$	§2.1.2
G, δ	the gross-exposure target the weights must sum in absolute value to, and the tolerance allowed on the net	§2.1.2
$\rho_s, \sigma_{MOM,s}, \sigma_{r,s}$	one asset’s signal-return correlation, and the standard deviations of its signal and its forward return	§2.1.2
N, i	lookback length, and the lag inside it running 1 to N	§2.1.1
x, y, β, ϵ	one observation’s signal value and forward return, the slope between them, and the part of the return the signal cannot reach	§2.2.3
ρ, R^2	their correlation, and its square—the share of the return’s variance the signal explains	§2.2.3
$\sigma_x, \sigma_y, \sigma_\epsilon$	standard deviation of the signal, of the return, and of the unreachable part	§2.2.3

Symbol	Means	First used
m, m_g	observations sharing a bucket, and the count in bucket g	§2.3.2
$\mu_g, \sigma_g, \text{SE}_g$	one bucket’s true mean forward return, the sample standard deviation of the returns in it, and the standard error of its mean	§2.3.2.3
H_0, z, α	the null that two buckets share a mean, the standardized G5 – G1 difference testing it, and the level it is tested at	§2.3.2.3
G1 ... G5	the buckets, lowest to highest signal	§2.3.2
$\sigma_{s,t}$	one asset’s volatility, estimated on data before that date	§2.4
R_p, r_f, σ_p	a portfolio’s return over a period, the financing rate subtracted from it, and its return volatility	§2.5
g	gap length—periods between the end of the signal’s window and the return it is scored on	§2.2.1

Note (Collisions to watch). α is the test level here, and `alpha` in code font is matplotlib’s opacity (§2.3.1); neither is 3’s smoothing constant. σ carries whatever its subscript says: σ_g is the spread of returns *inside* a bucket, SE_g the spread of that bucket’s mean, and $\sigma_{s,t}$ an asset’s trailing volatility—not 5 · §4.2’s σ_j , which rescales one signal. β is the slope of return on signal, never a market beta. g is the gap in §2.2.1 and a bucket index in §2.3.2.3; the subscript position tells them apart.

5 · Modelling

5.1 Why one split is not a backtest

5.1.1 The textbook split

Definition 5.1 (Split). A *split* cuts the sample by date into three contiguous blocks: **train**, on which coefficients are estimated; **validation**, on which choices between fitted models are made; and **test**, which is scored once and never chosen on.

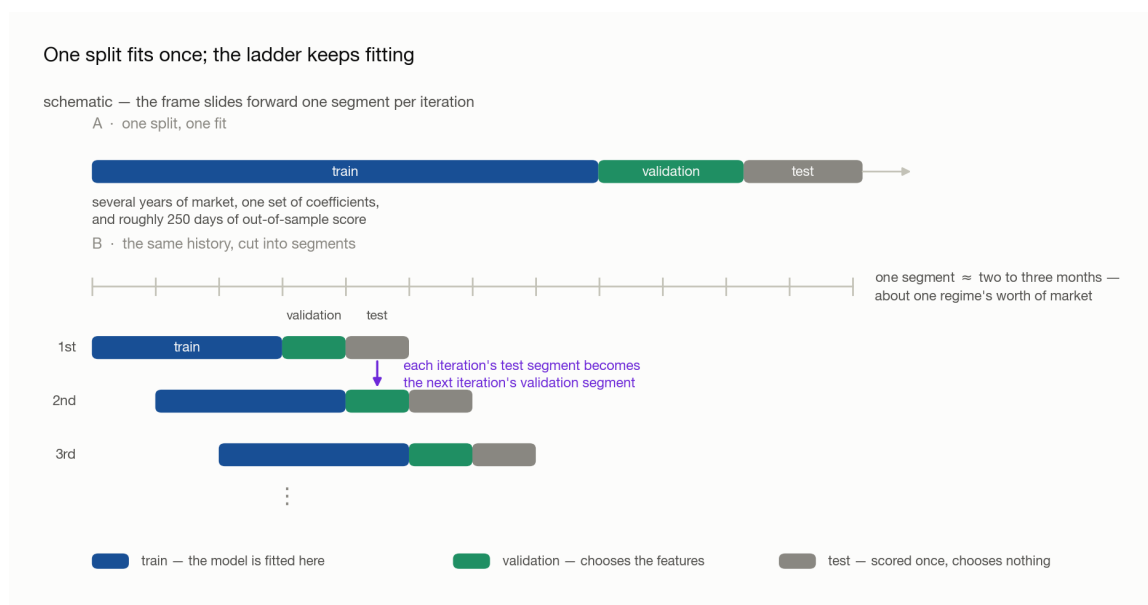


Figure 5.1: One split against the walk-forward ladder. Above, a single timeline cut once into train, validation and test. Below, the same history divided into twelve equal segments, with the train–validation–test frame starting one segment later on each row—so each row’s test segment is the next row’s validation segment.

The order matters and is not a convention: the blocks run forward in time, because a model that was fitted on next year and scored on last year has learned something no live book could have known.

5.1.2 What the single split costs

Three defects, and they compound.

Defect	What goes wrong
One fit discards time	A single set of coefficients over several years asserts that those years were one market. Early in the pandemic the world locked down at once and nobody wanted oil; the consensus was that demand had permanently moved online. It had not—the economy reopened, and energy then flew on war. A coefficient estimated across both states describes neither.
Almost nothing is left to score	Give validation a year and test a year and validation is roughly 250 dates. A year is also too little market: the regimes it contains are whichever ones happened to fall inside it, so the split never asks what the strategy does when the regime turns. A maximum drawdown measured over one calm year is not a risk estimate.
The sample is small and the signal is faint	A school dataset with a thousand points and an R^2 of 0.5 is ordinary, and in physics 0.9 is ordinary. In finance 0.1 to 0.2 is already a good R^2 —see 9. Weak true dependence plus few observations is precisely the regime in which a fit lands on noise: a line through ten points drawn from a cloud reaches R^2 0.8 easily.

5.1.3 The fix that is not one

Claim 5.2. Duplicating the sample changes no coefficient, and inflates every t-statistic by roughly $\sqrt{2}$.

Proof. Write the fit as $b = (X^T X)^{-1} X^T y$, with X the $n \times p$ matrix of signal values, y the forward returns, and b the estimated coefficients. Stacking the sample twice gives $X_2^T X_2 = 2 X^T X$ and $X_2^T y_2 = 2 X^T y$, so

$$b_2 = (2 X^T X)^{-1} 2 X^T y = b.$$

The residuals repeat as well, so the residual sum of squares doubles while the degrees of freedom go from $n - p$ to $2n - p$; for n large the variance estimate s^2 is therefore essentially unchanged. But the reported covariance is $s^2(X_2^T X_2)^{-1} = s^2(X^T X)^{-1}/2$ —halved. Standard errors fall by $\sqrt{2}$ and t-statistics rise by the same factor, on evidence that never arrived. $\square$

Note. This is the standard interview question, and the reason it is asked is that people reach for it. Manufacturing rows is overfitting with extra steps; the answer to too few observations is to score more of the ones you have, which is §5.2.

5.2 The walk-forward ladder

5.2.1 How long a segment should be

Definition 5.3 (Segment). A *segment* is a short contiguous block of dates—here **two to three months**—treated as one regime.

The length is set by how fast the market can change its mind, and the market is made of people. Within a day a regime shift is implausible: individuals change their view constantly, the crowd does not. Within a month, after a run of large news, one shift is plausible. Within a year or two there is room for several, because that is enough time for the same participants to change their minds repeatedly. So a segment must be short enough that one segment is approximately one regime, and long enough to hold a usable sample.

5.2.2 The slide

Cut the history into segments 1, 2, 3, . . . and slide a frame across them:

Iteration	Train	Validation	Test
1	1–3	4	5
2	2–4	5	6
3	3–5	6	7

Claim 5.4. The ladder repairs all three defects of §5.1.2 at once.

Each fit now sees one regime’s worth of data and is re-estimated as the market moves on, so time is no longer averaged away. Every segment past the first frame is eventually scored out of sample, so the out-of-sample record grows with the length of the history instead of being fixed at one year. And a regime shift that lands inside a validation segment is now a *test* the strategy has to pass, rather than an event the single split happened to miss.

Note. Read the table down its two right columns: **each iteration’s test segment is the next iteration’s validation segment.** Nothing is discarded between rungs, and no segment is ever scored twice under the same coefficients.

5.2.3 One rung, in full

A rung is not a single fit. Train ends, a validation segment intervenes, and only then does test begin—so a model fitted on train alone is asked to predict across a gap of market it has never seen, and pays for it.

The repair is the two-step shape that Lasso already imposes. Lasso adds an L_1 penalty to the least-squares objective,

$$\min_{\beta} \sum_t \left(y_t - \sum_j \beta_j x_{j,t} \right)^2 + \lambda \sum_j |\beta_j|$$

where y_t is the forward return on date t , $x_{j,t}$ signal j 's value on that date, β_j its coefficient, and $\lambda \geq 0$ the penalty strength. The penalty drives coefficients exactly to zero, which makes it a **feature selector**; and because the surviving coefficients are biased toward zero by that same penalty, the standard practice is to **refit** the survivors without the penalty before predicting.

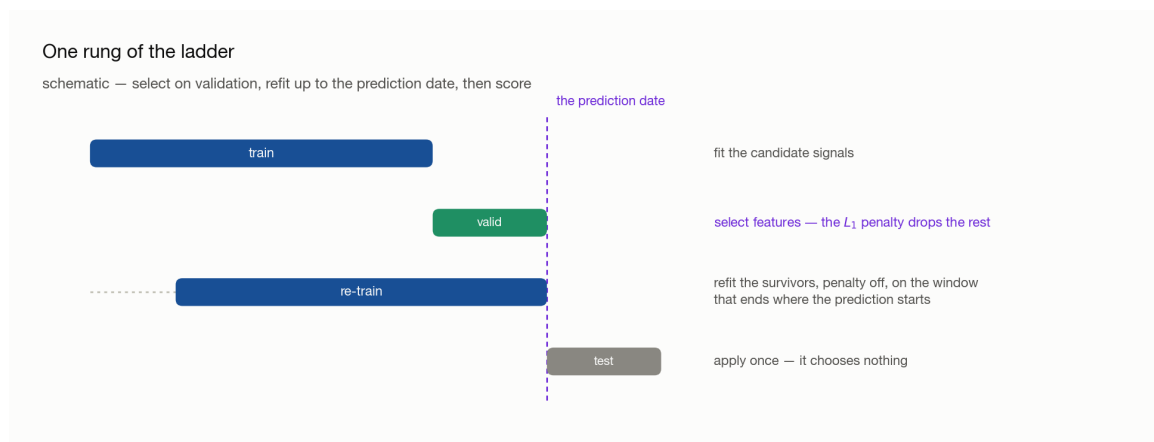


Figure 5.2: One iteration, in four stages: fit on train; select features on the validation segment; refit the survivors on a window shifted forward so that it ends where the prediction starts; apply once to test, beyond the dashed prediction date.

So each rung runs: **fit on train**, **select features on the validation segment**, **refit the survivors on a window ending where the prediction starts**, **apply once to test**.

Note (Two house styles). The refit window is either the training window shifted forward, or the training window plus the validation segment. The sample sizes differ little and both are used in industry; it is a preference, not a correctness question. What is not optional is that the window ends at the prediction date—see §5.4.3.

5.3 Two validation layers

The validation *segment* inside each rung and the large held-out validation *block* at the end of the history are different objects with different powers.

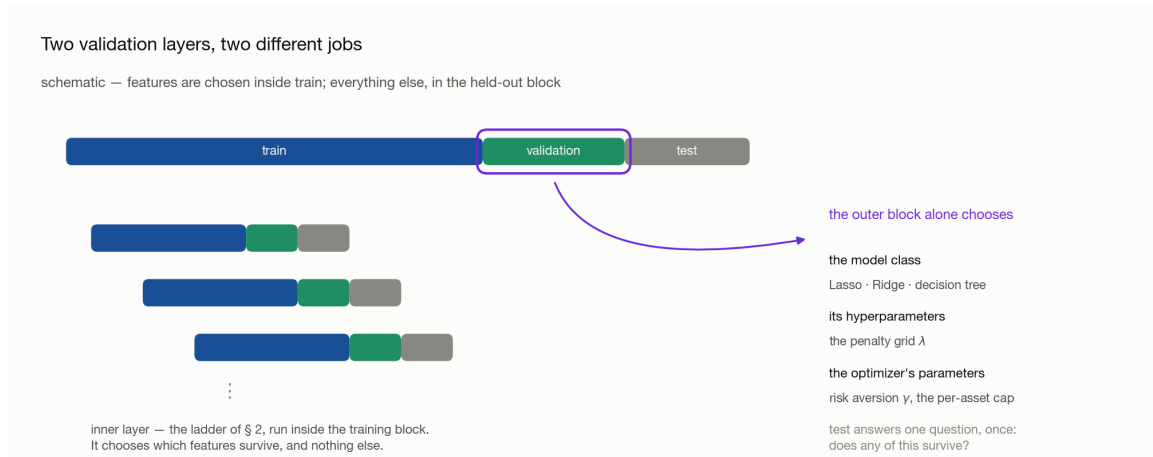


Figure 5.3: The two layers. The inner layer is the miniature ladder inside the training block, which chooses which features survive; the outer layer is the ringed held-out validation block, which chooses the model class, its hyperparameters, and the optimizer's parameters.

Layer	May choose	May not
Inner —the validation segment inside each rung	which features survive, and the refit that follows	anything held constant across rungs
Outer —the held-out validation block	the model class , its hyperparameters (λ), and the optimizer's parameters	coefficients—those are fitted on train
Test	nothing	everything

5.3.1 Why the inner layer cannot be skipped

Exploration already looked at which signals worked *in the training period*. Of course the fit then describes that period well: momentum was selected partly because it was seen to work there. That says nothing about whether it was useful going in, and the only way to find out is to select on dates the selection did not see—which is what the validation segment is for.

Note. A signal's usefulness is not a constant either. When more signals enter the book, or when one starts to fail, momentum should no longer carry the weight it carried alone. Because the ladder re-selects at every rung, that adjustment happens on its own rather than being asserted once.

5.3.2 What only the outer block may decide

λ is not fitted, it is searched: run the ladder once per value on a grid—0.1, 0.5, 0.7, 1—and compare. The same is true of the model class itself (Lasso, Ridge, a decision tree). Both comparisons need dates that no rung of the ladder consumed, which is exactly what the held-out block is.

5.3.3 The optimizer's parameters

The prediction is not yet a portfolio. Weights come from a mean–variance problem: choose w to

$$\max_w E[r_p] - \gamma \cdot \text{Var}[r_p]$$

subject to constraints such as

$$\sum_i w_i = 1.0 \quad \sum_i |w_i| \leq 2.0 \quad |w_i| \leq 0.20$$

—net exposure 100 percent, gross exposure at most 200 percent, and no single asset above 20 percent—where w_i is the weight on asset i , $E[r_p]$ and $\text{Var}[r_p]$ the portfolio's expected return and variance, and $\gamma \geq 0$ the **risk-aversion coefficient**.

Two of these are free parameters, and neither can be reasoned to:

- γ . Nobody can say from first principles whether a risk appetite is 0.3, 0.5 or 1. The number has no interpretable units—it is a trade rate between two quantities measured in different things—so it is chosen by running the grid and reading the result.
- **The per-asset cap.** With 37 assets, a 20 percent cap may be too tight for the book to hold enough of whichever names are paying; 25 or 30 percent may be the right ceiling. Gross and net exposure, by contrast, are set by mandate and are not tuned.

Note. All of this tuning belongs in the **outer** block. Tuning γ inside the rungs would let a portfolio parameter be chosen on the same dates the features were chosen on, and the held-out block would then be measuring a strategy that had already been fitted to it.

5.4 Preparing a signal for a model

Two transformations stand between a raw signal and a model, and both exist to stop the fit from tracking things that are not the market.

5.4.1 Clipping: keep the observation, cap the value

Claim 5.5. A ranked signal does not fill its range evenly—the mass piles at both ends—and those ends move the fit far more than their count suggests.

The first half is empirical: momentum in particular pins at an extreme whenever a name runs, so the histogram is a barbell rather than a plateau. The second half is mechanical. In a least-squares fit each observation's pull on the slope grows with its distance from the mean of x , so a handful of far points can swing a coefficient that

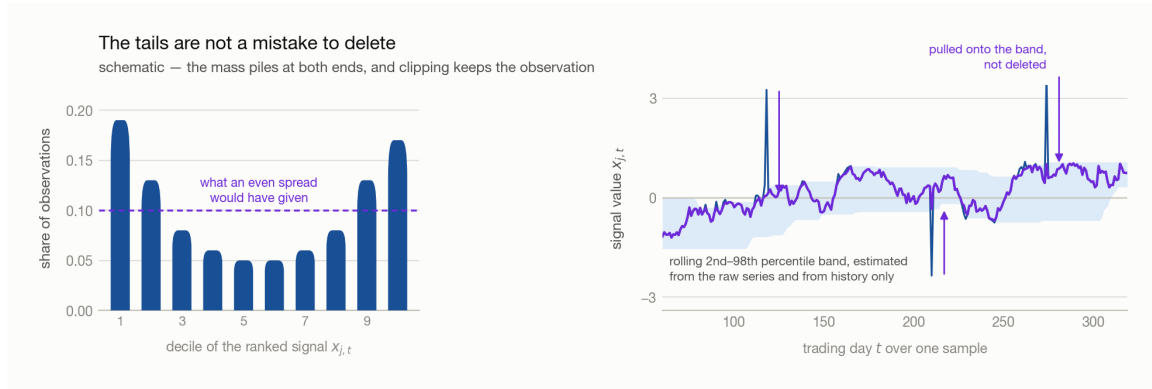


Figure 5.4: Left, the share of observations by decile of the ranked signal—high at the first and last deciles, low in the middle, against the dashed line at 0.1 that an even spread would give. Right, a signal series against a shaded rolling percentile band, with the spikes that reach outside it pulled back onto its edge.

hundreds of central points barely move. The coefficient then jumps for reasons that have nothing to do with the market.

Deleting the far points, which is the textbook move, fails twice here:

- **The sample is already small.** Every deletion sharpens §5.1.2’s third defect, and in a messy sample it is not even clear which points are outliers.
- **The tail carries information.** What the book does when a signal goes extreme is exactly what the model should learn. Delete those dates and it never can.

Definition 5.6 (Winsorization). Replacing any value outside a chosen quantile band by the nearer edge of the band:

$$x_{j,t}^{\text{clip}} = \min \left(\max \left(x_{j,t}, Q_{j,t}^{\text{lo}} \right), Q_{j,t}^{\text{hi}} \right)$$

where $Q_{j,t}^{\text{lo}}$ and $Q_{j,t}^{\text{hi}}$ are quantiles of signal j over the trailing L dates—1 and 99 percent, or 2.5 and 97.5 percent; there is no canonical pair.

Note (Compare raw to raw). The band for date t must be estimated from the **un-clipped** history. Feed yesterday’s capped value into today’s quantile and the band walks inward on itself, a little further every step. Keep the raw column and produce the clipped one beside it.

Note (Why the band rolls). A value that is extreme against the last three months may be unremarkable six months later, once the market has moved to it. A fixed threshold pins such a signal at the 99th percentile forever; a rolling one releases it as soon as the surrounding distribution catches up.

→ `rolling`, `quantile` and `clip` as pandas calls: 8 · Toolbox.

5.4.2 Standardizing: put every signal on one scale

Momentum lives in roughly $[-0.05, 0.05]$. VIX lives in $[0, 100]$. Fit both and the coefficients absorb the mismatch.

Claim 5.7. Coefficient magnitudes carry no information about a signal’s importance until the signals share a scale.

Proof. Replace x_j by $c x_j$ for some $c > 0$. The fitted values are unchanged if β_j is replaced by β_j/c , and least squares chooses fitted values—so the fit is identical and the coefficient is whatever the units make it. A comparison of β magnitudes across differently scaled signals is therefore a comparison of units. $\square$

The consequence is not merely aesthetic. Every input is an **estimate** and carries error: a momentum measured from returns is the residue of participants entering and leaving, and a jump in it is as likely to be noise as news. A large β multiplies that noise into the prediction, while the small β on the wide-ranging signal cannot respond when it genuinely moves.

Definition 5.8 (Standardized signal).

$$z_{j,t} = \frac{x_{j,t} - \mu_j}{\sigma_j}$$

with μ_j and σ_j the mean and standard deviation of signal j over the estimation window. Dividing by a rolling standard deviation, or by a cross-sectional sum, are alternatives; what matters is that **every signal is treated the same way**.

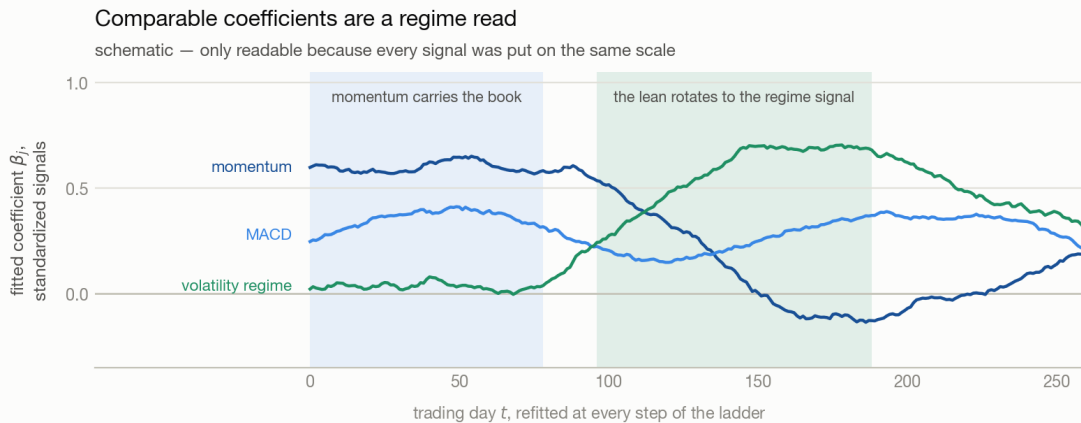


Figure 5.5: Three coefficient paths over a year of refits: momentum decays from high toward zero, a volatility-regime signal rises from zero to become the largest, and MACD stays between them. The shaded stretches mark where momentum carries the book and where the lean rotates to the regime signal.

Three things follow, in increasing order of usefulness:

Consequence	Why
Coefficients become comparable	With units removed, β_j measures response per standard deviation of signal—the same question for every signal.
The coefficient path becomes a regime read	Plot every β_j from every rung on one axis. Momentum dominating one stretch and volatility another is the change in market sentiment, in a form you can look at.
Errors partly cancel	On one scale, with comparable coefficients and errors that are close to independent across signals, the noise in the combined prediction averages down as signals are added, instead of being dominated by whichever signal has the largest units.

5.4.3 Which mean and standard deviation you may use

μ_j , σ_j and the quantile band all have to be estimated from somewhere, and that choice is where look-ahead enters.

The rule follows from **where the model is built**: at the *end* of the training window. Every date inside that window is history at that moment, so standardizing with the whole training window's mean and standard deviation is legitimate—even though, from the point of view of an individual date early in the window, the statistic used to scale it was computed partly from its own future.

At this point of construction	These statistics are available
End of the training window	the whole training window
End of the training window	never the validation segment, and never test
End of a refit window	the whole refit sample, validation segment included

Note. The distinction is between *a signal's* information set and *the model's*. A signal computed on date t may use only data up to t , because it has to be computable live. The model is not computed on date t ; it is computed once, at the end of its window, and then applied forward. Confusing the two costs either a leak or an estimate worse than the one you were entitled to.

Note. This is the failure worth being paranoid about. In a time series the smallest leak makes the result beautiful, and the gap between a beautiful backtest and the live book is where the money goes.

Appendix · Notation

Throughout, t is the date, i indexes assets and j indexes signals.

Symbol	Means	First used
$x_{j,t}, x_{j,t}^{\text{clip}}, z_{j,t}$	signal j on date t : raw, after clipping, after standardizing	§5.2.3, §5.4.1, §5.4.2
y_t, β_j	the forward return being predicted, and the coefficient the fit puts on signal j	§5.2.3
X, y, b, n, p, s^2	the design matrix, the response, the estimated coefficients, the number of dates, the number of signals, the residual variance estimate	§5.1.3
λ	the L_1 penalty strength—searched on a grid, not fitted	§5.2.3
L	the trailing window, in dates, that the clip band is estimated over	§5.4.1
$Q_{j,t}^{\text{lo}}, Q_{j,t}^{\text{hi}}$	the low and high quantiles of that window	§5.4.1
μ_j, σ_j	the mean and standard deviation used to standardize signal j	§5.4.2
$w_i, \gamma, E[r_p]$	the portfolio weight on asset i , the risk-aversion coefficient, and the portfolio's expected return	§5.3.3

Note (Collisions to watch). σ_j here is one **signal's** dispersion, used only to rescale it—not 2 · §4's $\sigma_{s,t}$, an asset's trailing volatility, and not 4 · §1.1's σ , the amplitude of the tape. β_j is a coefficient on a **signal**, as in 4 · §3.2, never a market beta. λ is the penalty and γ the risk aversion: the lecture wrote both as λ and then renamed the second, and they are unrelated. L is a trailing window as in 4 · §2.1, here over a signal rather than over returns.